NLP 243 Machine Learning for Natural Language Processing

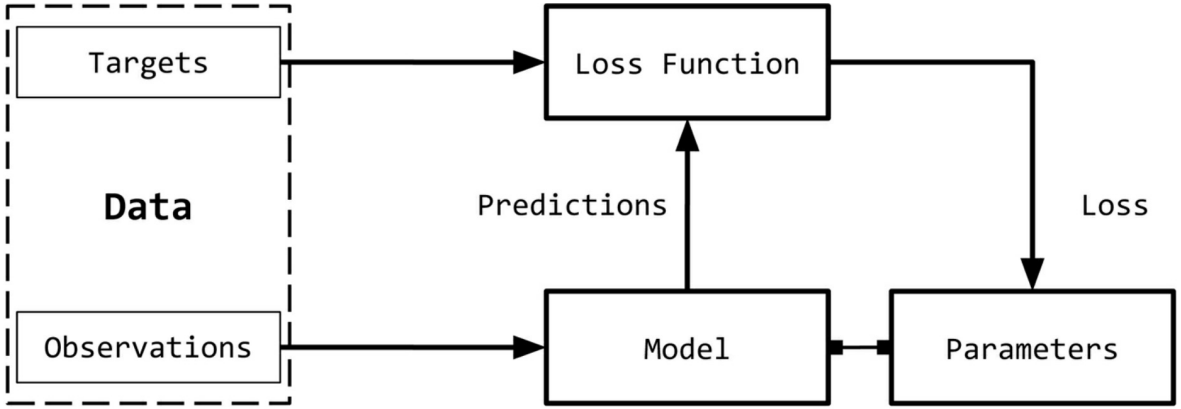
Fall 2023

Outline for Today

- Some Terminology
- What is deep learning?
- Probability
- Conditional Probability and Independence

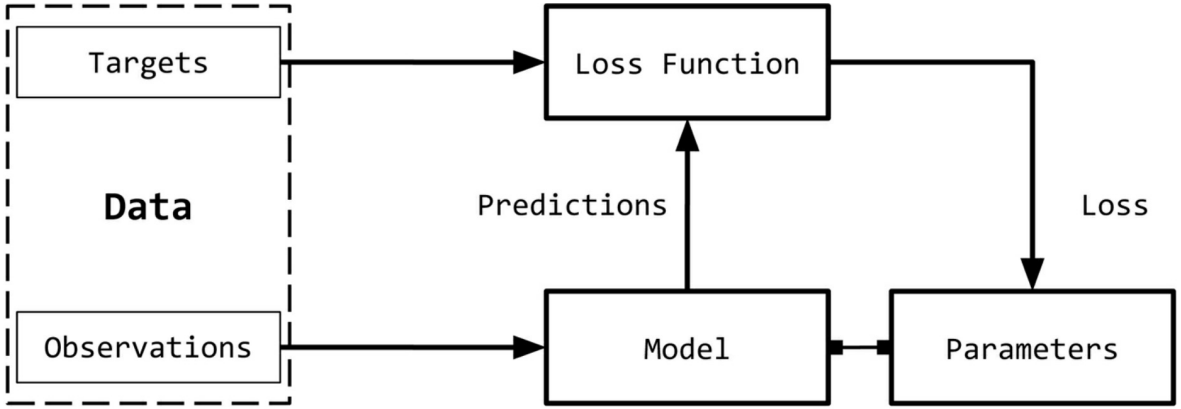
Terminology

- **Observations**: inputs, x.
- **Targets**: labels, y, corresponding to the observations (sometimes called "ground truth").
- **Model**: Function that takes an observation and predicts the value of its target label.



More Terminology

- **Parameters**: weights (for deep learning), w, that parameterize the model.
- **Predictions**: Also called "estimates", target values estimated by the model, $\hat{y}$.
- **Loss Function**: a function (denoted $\mathcal{L}$) that compares how far off a prediction is from its target. The lower the loss, the better is the model at predicting the target.



Supervised Learning - Regression

- When the targets take arbitrary values in a range (instead of specific categories such as +/- for sentiment classification), we call it a **regression** problem.
- Examples:
  - Estimating the price of a house given attributes such as location, square footage, number of rooms, etc.
  - Predicting how many miles can a dog run given attributes such as its breed, age, size, etc.

Supervised Learning - Classification

- When the targets take a limited number of pre-defined categories, we call it a classification problem.

- **Binary classification:** only two values
  - Positive/negative sentiment classification
  - Sentence segmentation from speech: each word boundary is classified into sentence boundary versus not a sentence boundary.

- **Multi-class classification:** more than two values
  - Digit recognition: each digit is classified into one of 10 values for 0,…,9.

Supervised Learning – Tagging

- Some tasks are not simply binary or multi-class classification tasks, but we need to
  - Tagging images with multiple objects
  - Tagging tokens/words in natural language utterances

Locations in utterances:
I want to fly from Boston to Seattle via Chicago. =>
I want to fly from Boston to Seattle via Chicago.

Sequence Tagging

Suggestions and Readings

- NLP243 Section Tomorrow at 1:30pm RM2225
  - Get started with PyTorch, Colab and Homework 1

		Release Date	Due Date
In-class Quizzes			
Homework 1		Tuesday, October 3rd	Monday, October 16th
Homework 2		Tuesday, October 17th	Monday, October 30th
Homework 3		Tuesday, October 31st	Monday, November 13th

- Readings:
  - Ch1-2 of NLP with PyTorch
  - Ch1-4 of Dive into Deep Learning

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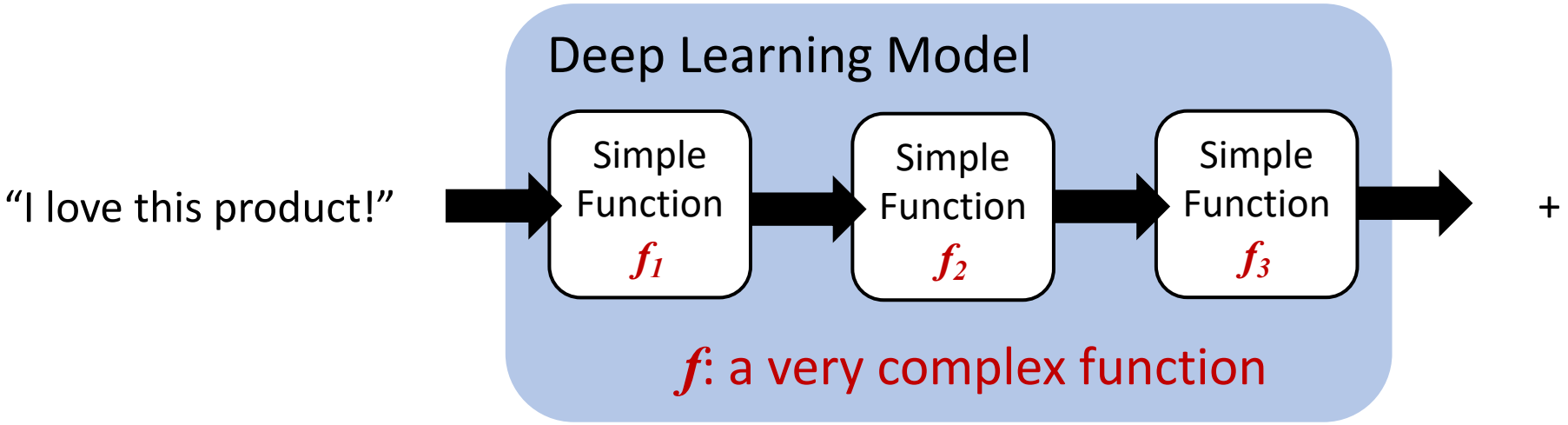
History of Deep Learning

- 1960s: Perceptron (single layer neural network)
- 1969: Perceptron has limitations
- 1980s: Multi-layer perceptron
- 1986: Backpropagation
- 1989: 1 hidden layer is “good enough”, why deep?
- 2006: RBM initialization *
- 2009: GPU
- 2010: Speech Recognition * (Dahl et al., 2010)
- 2012: ImageNet * (Krizhevsky et al. 2012)
- 2015: “superhuman” results in Image and Speech Recognition

Stacked Functions Learned by Machine

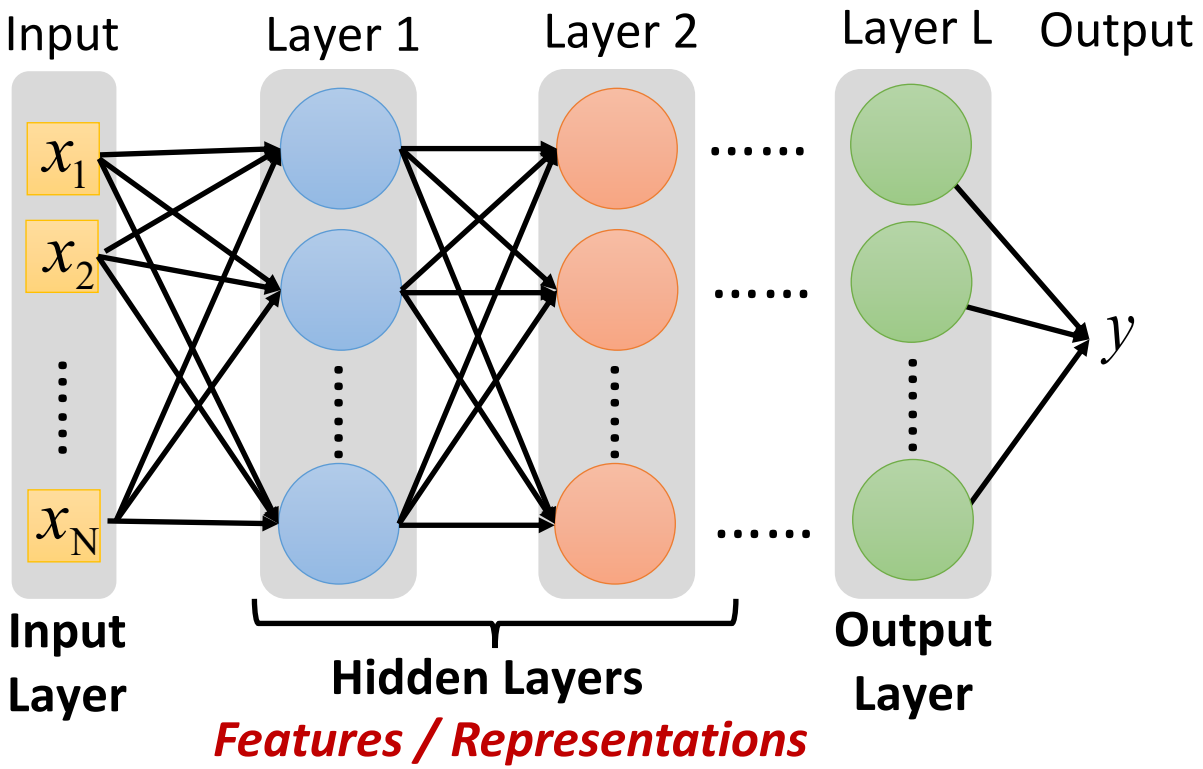


End-to-end training: what each function should do is learned automatically

Deep learning usually refers to *neural network* based model

Stacked Functions Learned by Machine

vector $\boldsymbol{x}$
"I love this product."



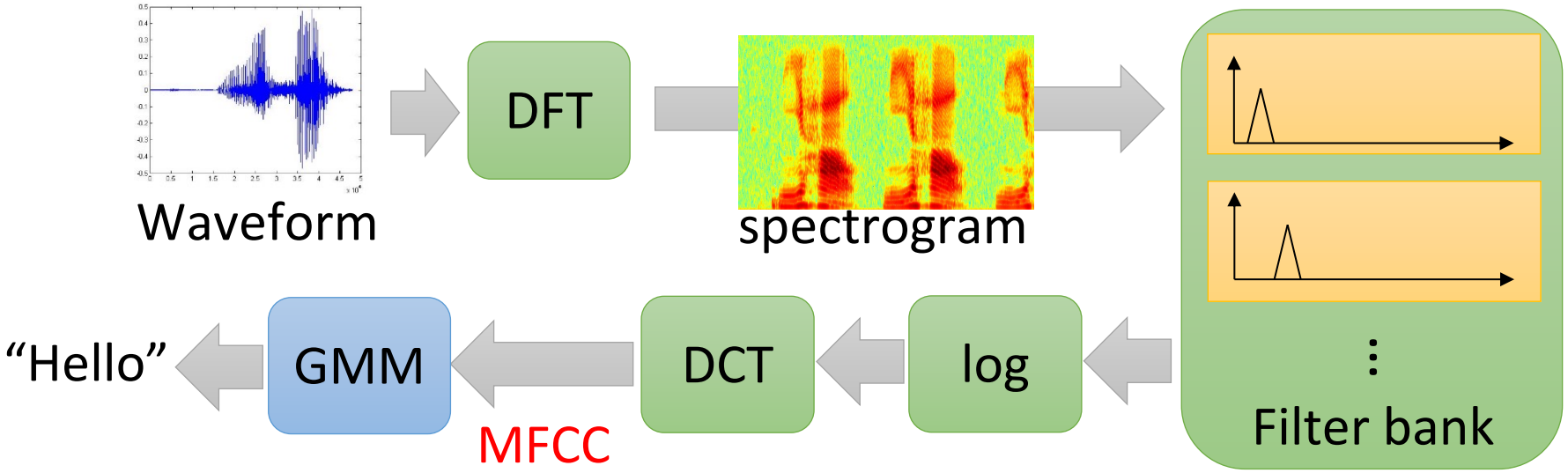
label $\boldsymbol{y}$
+

Representation Learning attempts to learn good features/representations

Deep Learning attempts to learn (multiple levels of) representations and an output

Deep vs. Shallow – Speech Recognition

- Shallow Model

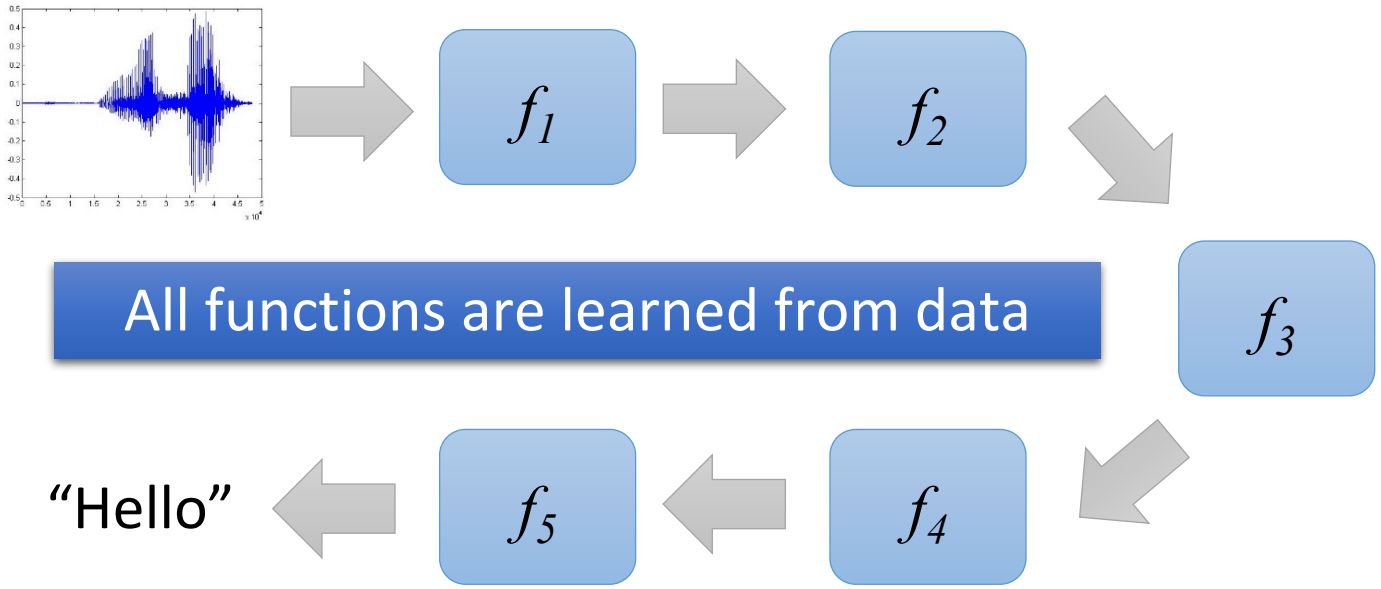


Each box is a simple function in the production line:

:hand-crafted :learned from data

Deep vs. Shallow – Speech Recognition

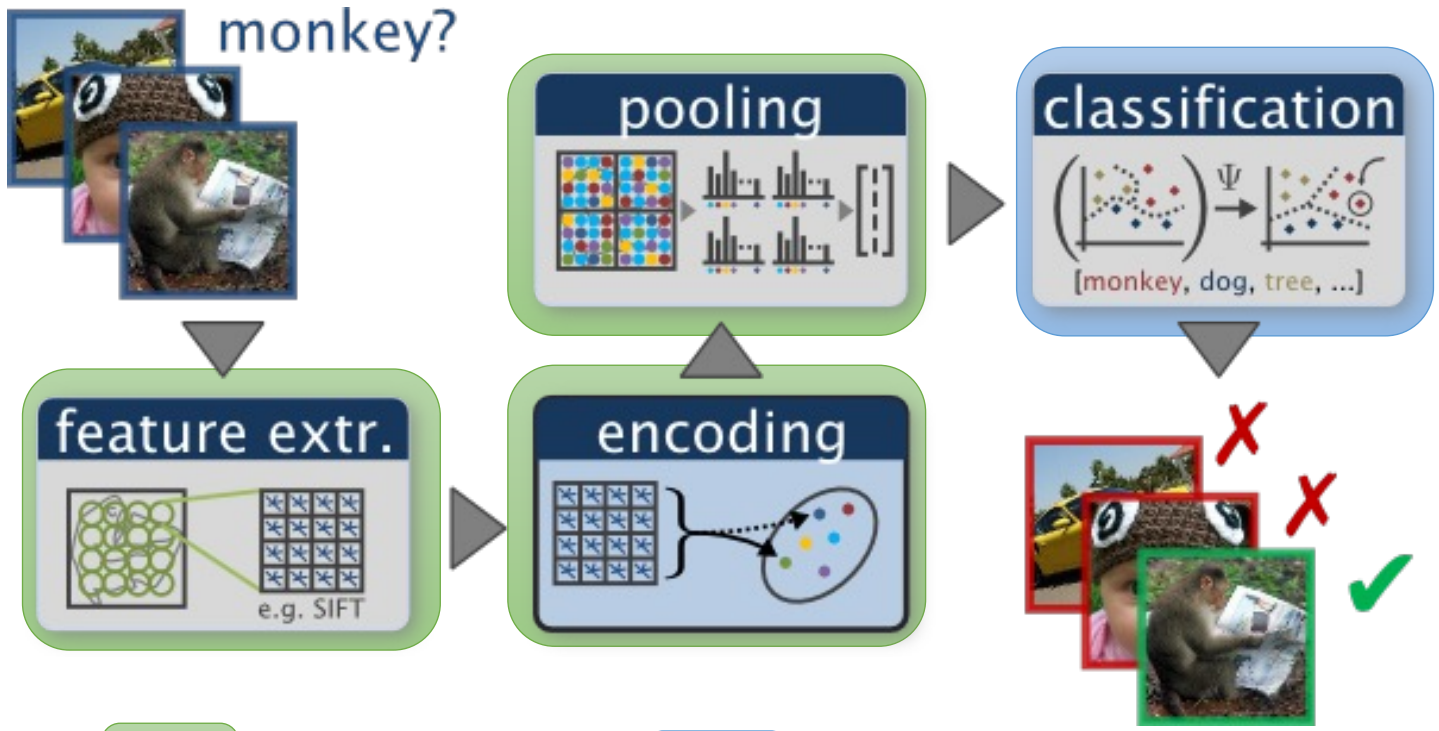
- Deep Model



Less engineering labor, but machine learns more

Deep vs. Shallow – Image Recognition

- Shallow Model



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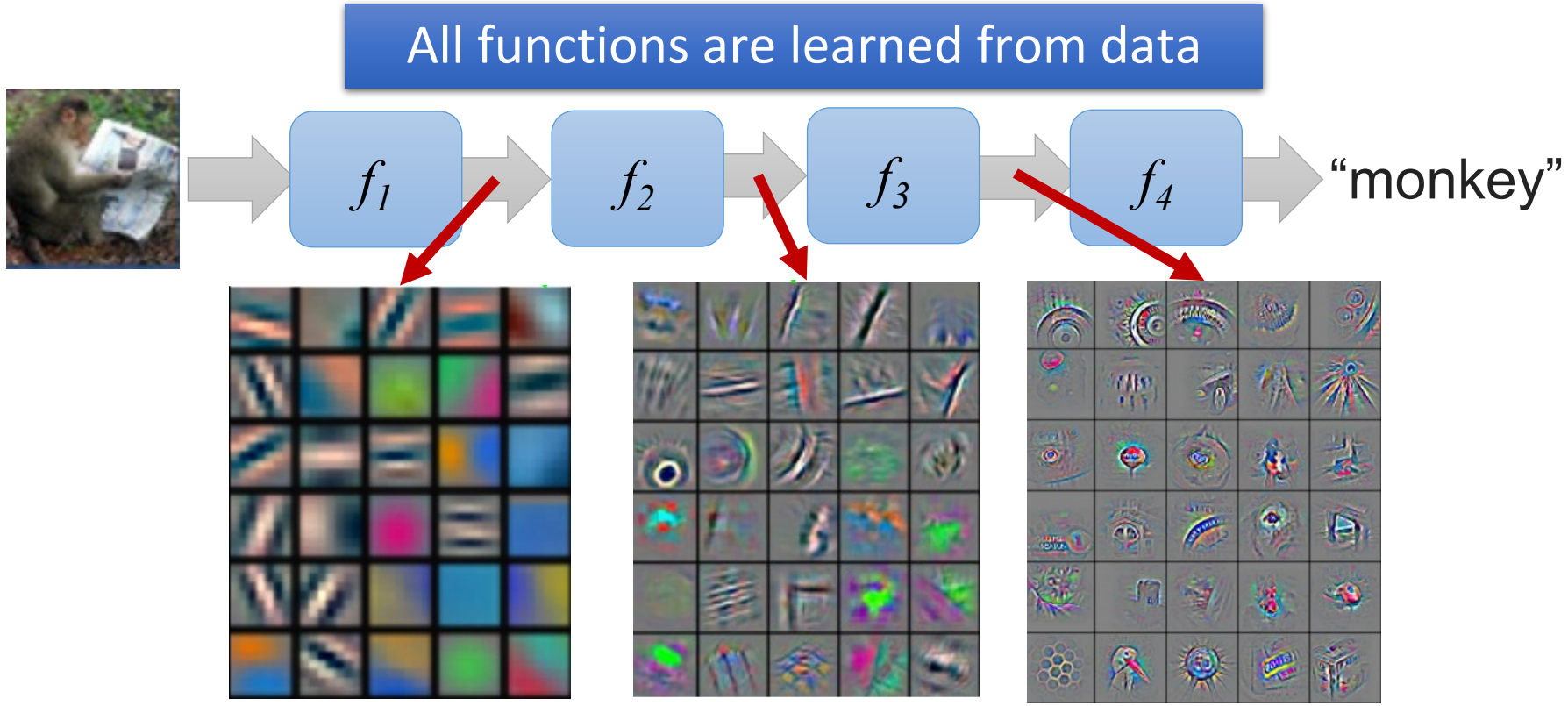
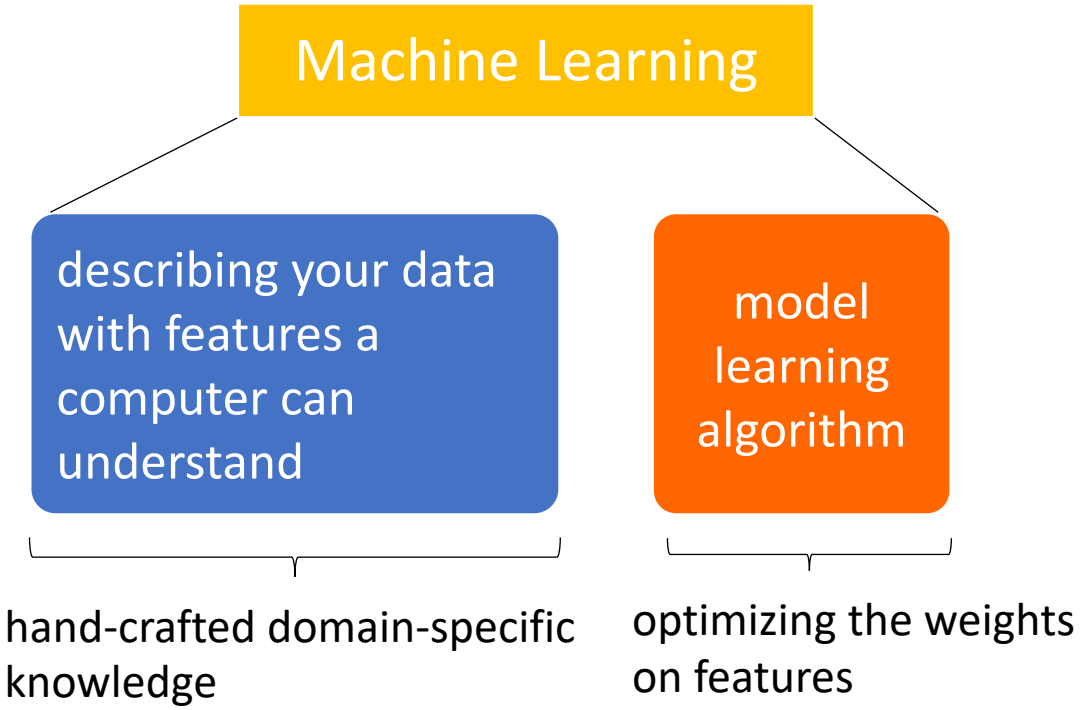
Deep vs. Shallow – Image Recognition

- Deep Model

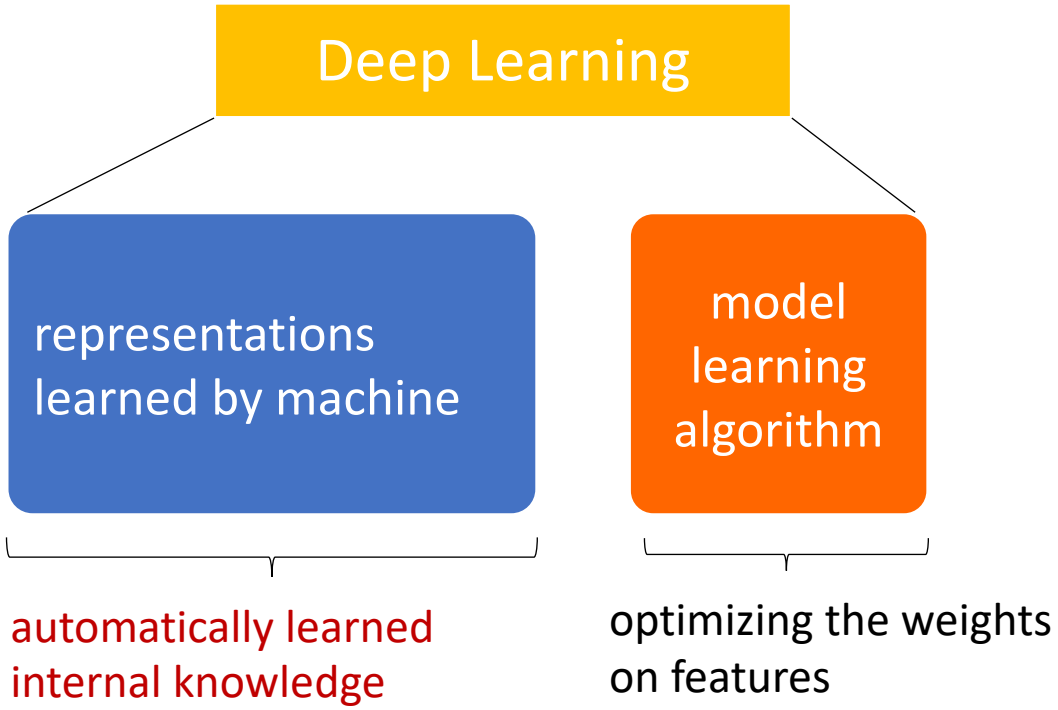


Features / Representations

Machine Learning vs. Deep Learning



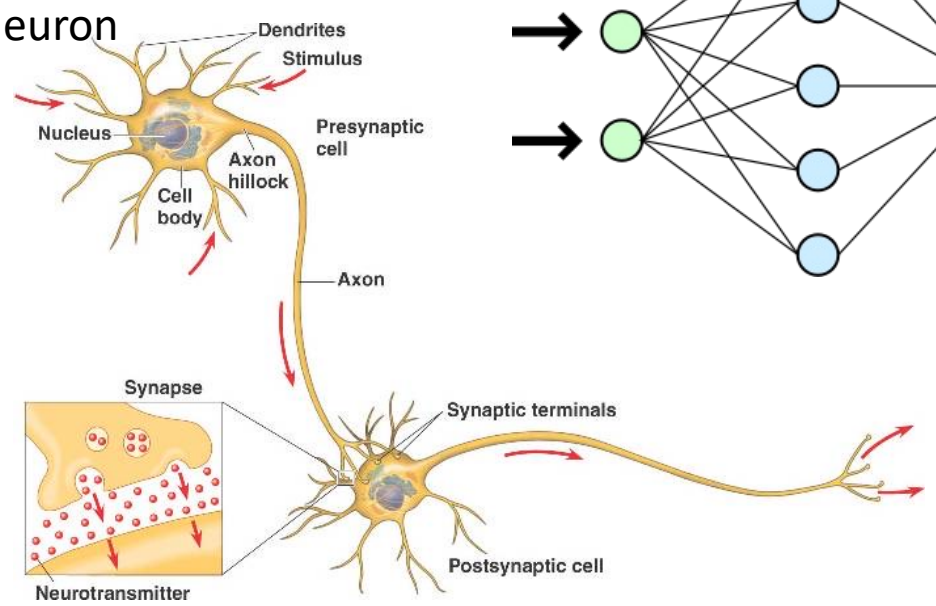
Machine Learning vs. Deep Learning

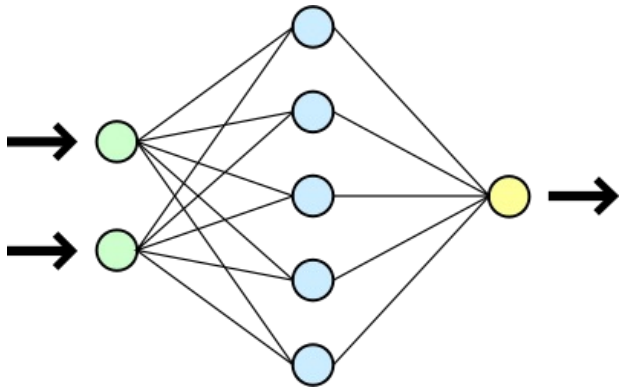


Deep learning usually refers to *neural network* based model

Inspired by Human Brain

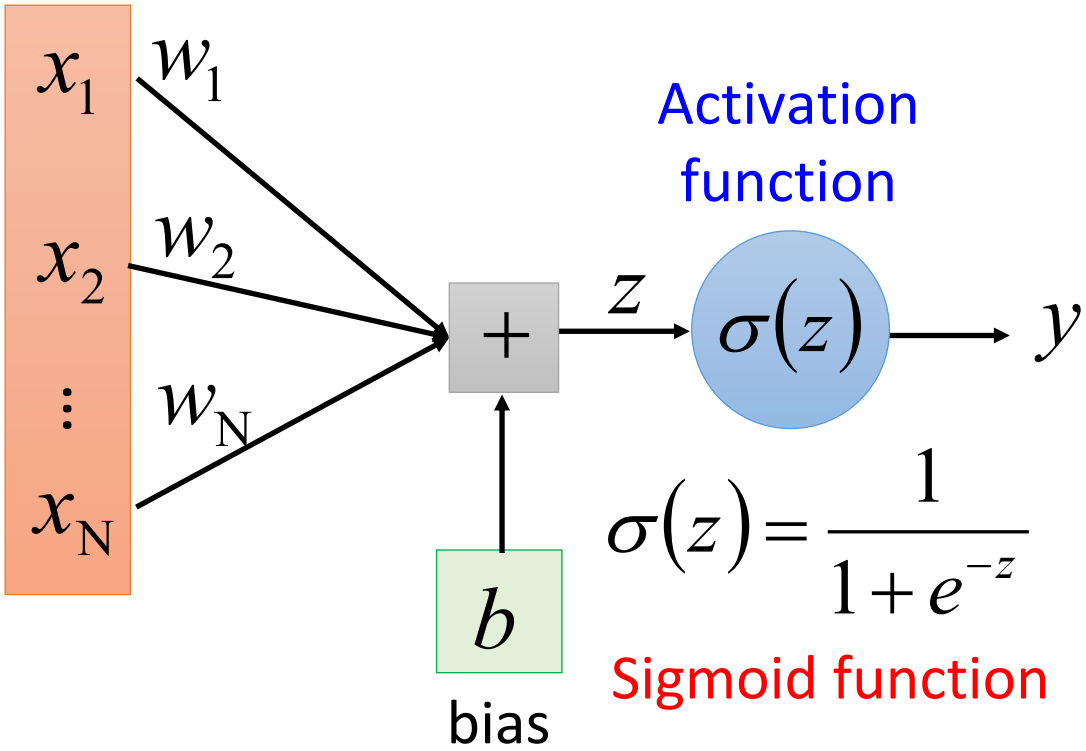
neuron





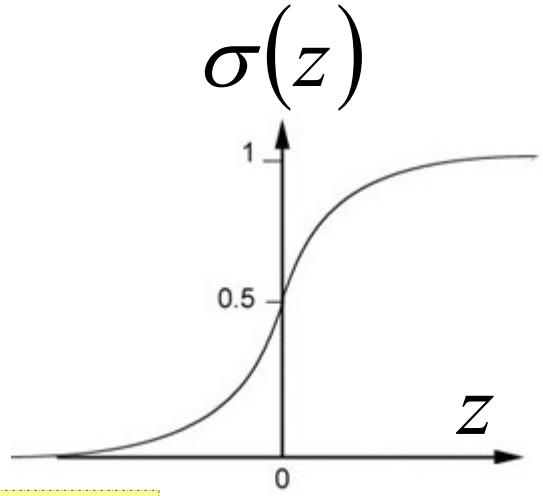
Neurotransmission section of: https://en.wikipedia.org/wiki/Human_brain

A Single Neuron



$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Sigmoid function



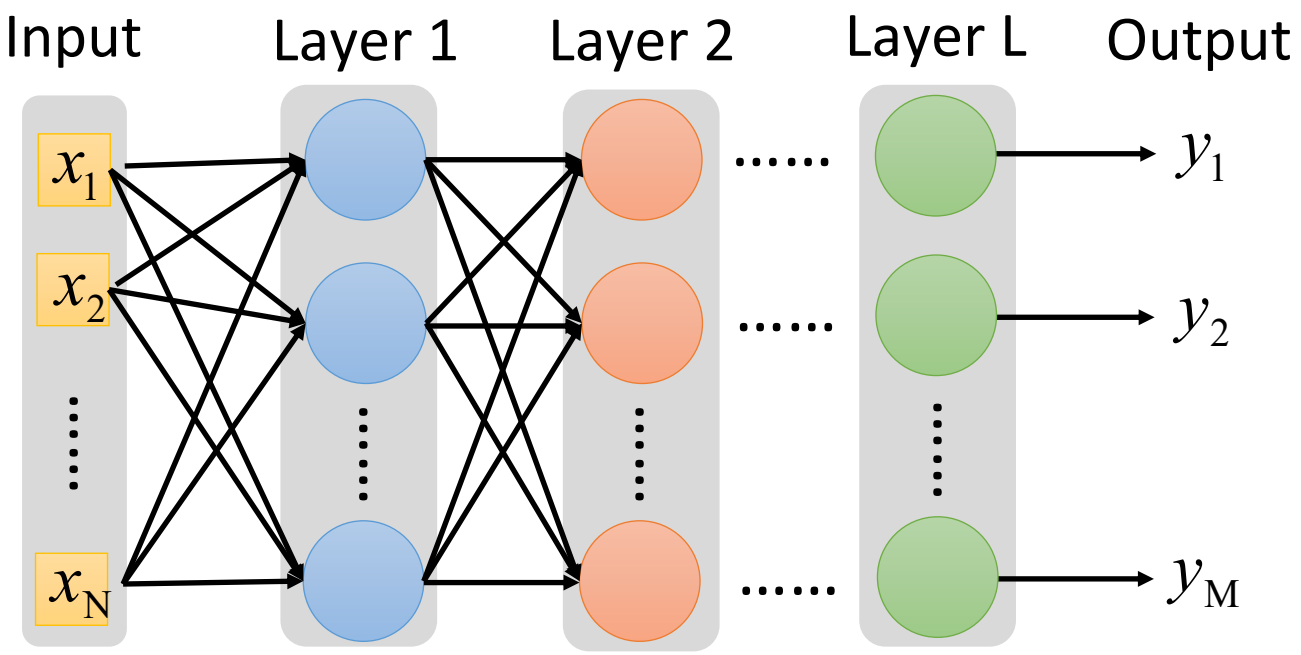
Each neuron is a very simple function

Deep Neural Network

A neural network is a complex function:

$$f : R^N \rightarrow R^M$$

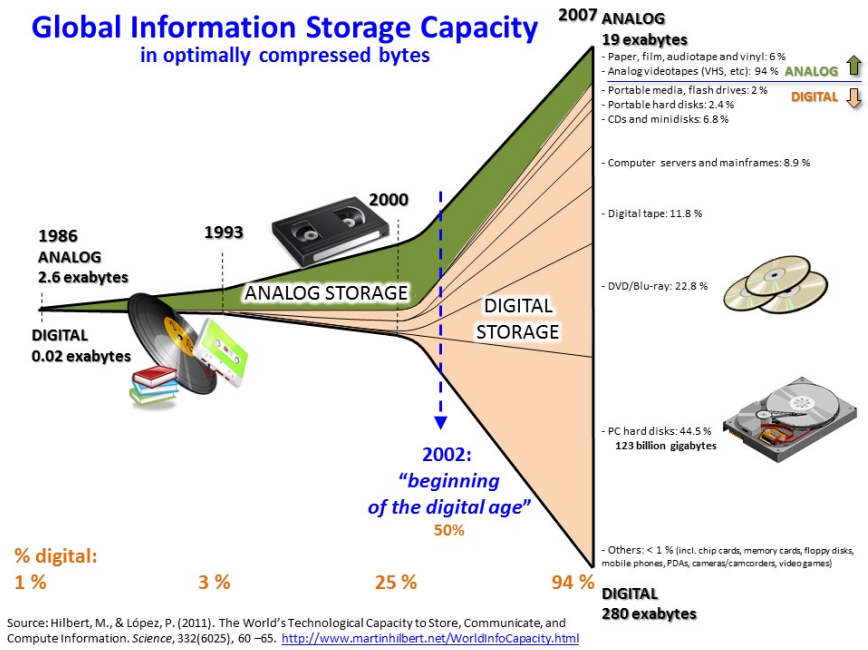
- Cascading the neurons to form a neural network



Each layer is a simple function in the production line

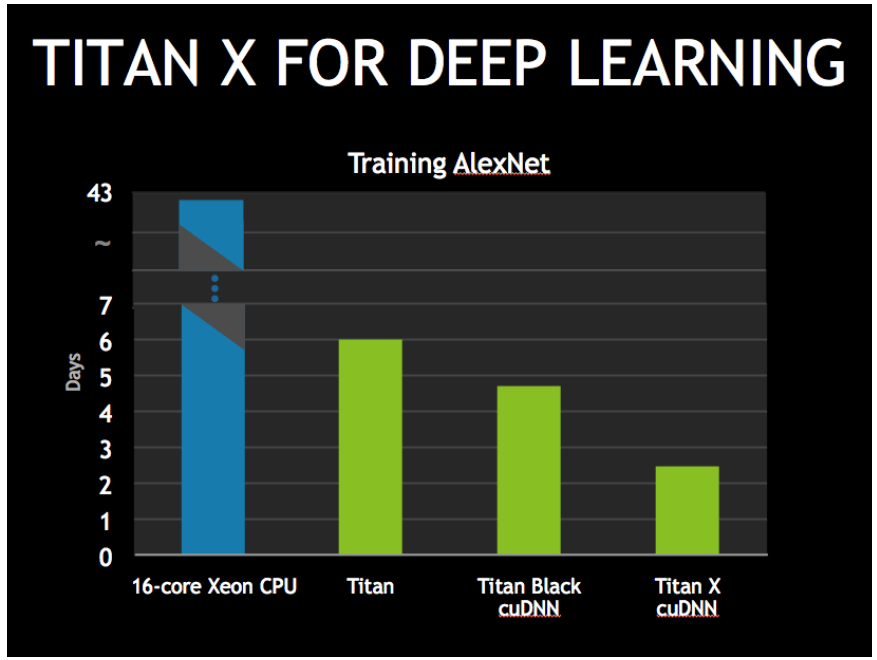
Why does deep learning show breakthrough in applications after 2010?

- Big Data



Source: Hilbert, M., & López, P. (2011). The World's Technological Capacity to Store, Communicate, and Compute Information. *Science*, 332(6025), 60 –65. http://www.martinhilbert.net/WorldInfoCapacity.html

- GPU

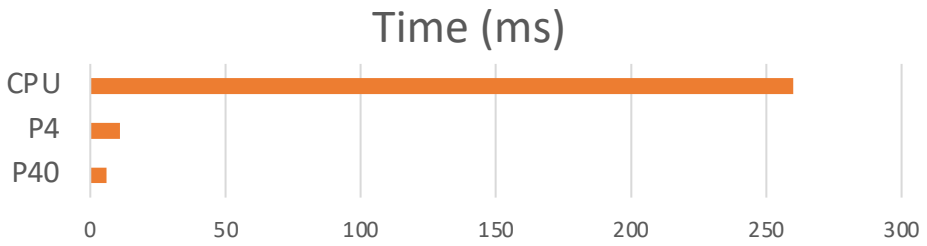


And algorithmic improvements!

Why does speed matter?

- Training time
  - Big data increases the training time
  - Too long training time is not practical

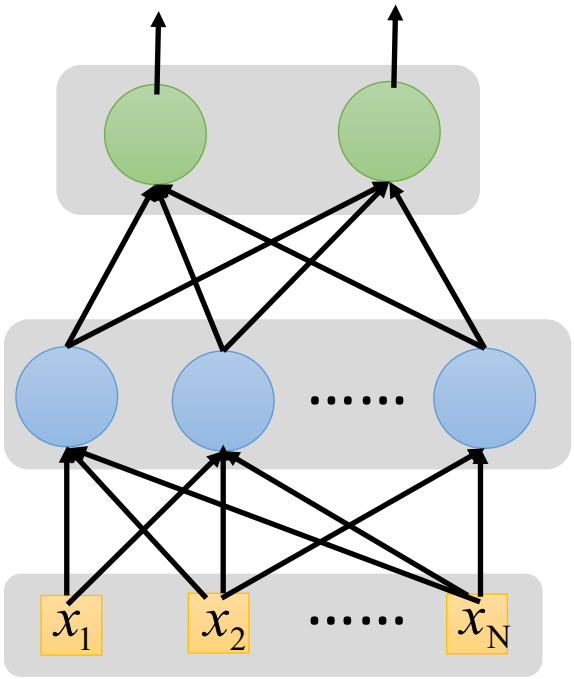
- Inference time
  - Users are not patient to wait for the responses



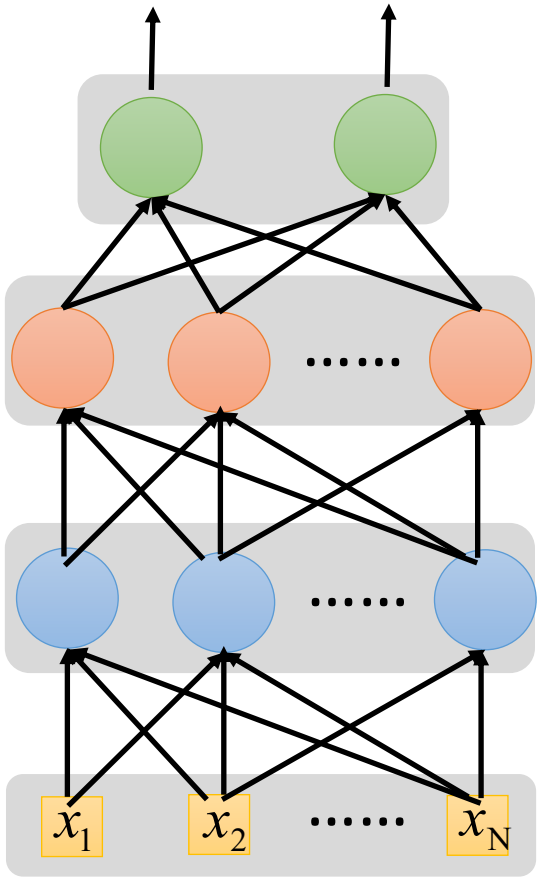
GPU enables the real-world applications using the computational power

Why is deeper better?

- Deeper → More parameters?



Shallow



Deep

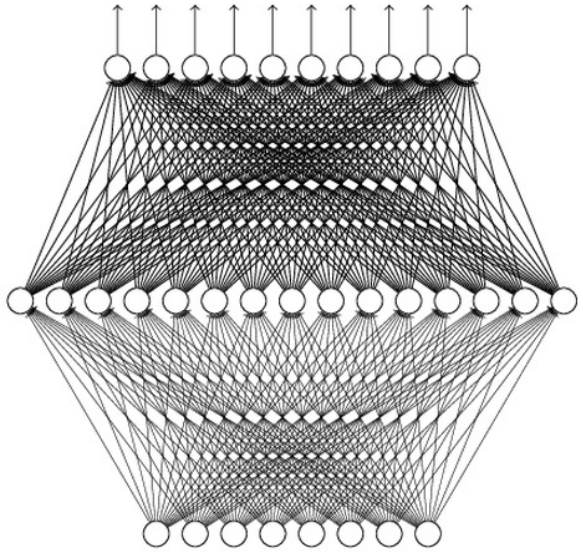
Universal Approximation Theorem

- Any continuous function f

$$f : R^N \rightarrow R^{\mathrm{M}}$$

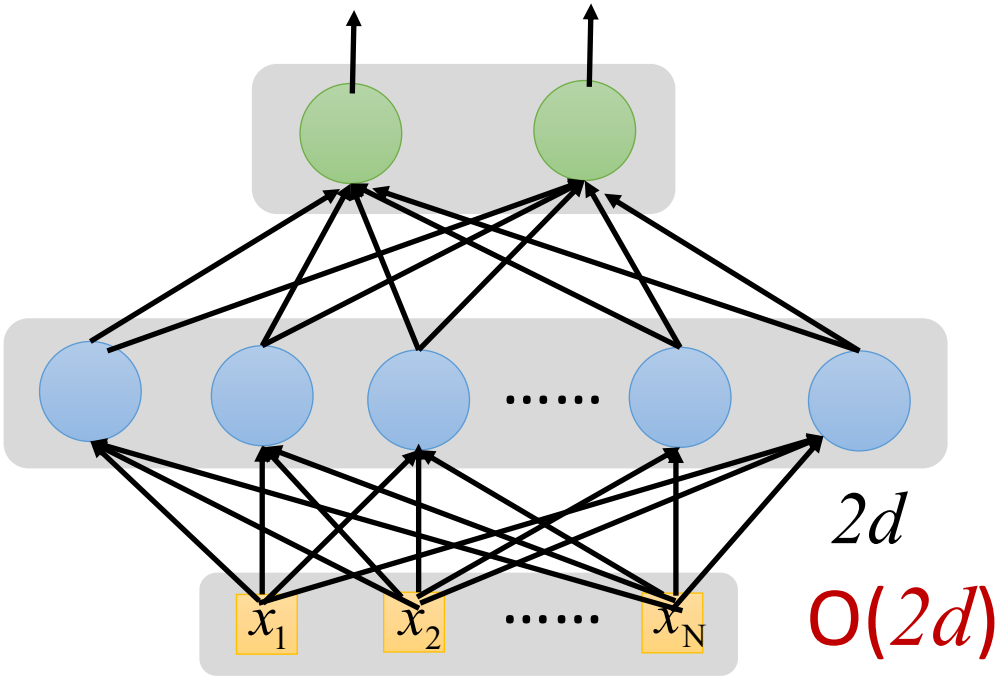
- Can be realized by a network with only one hidden layer

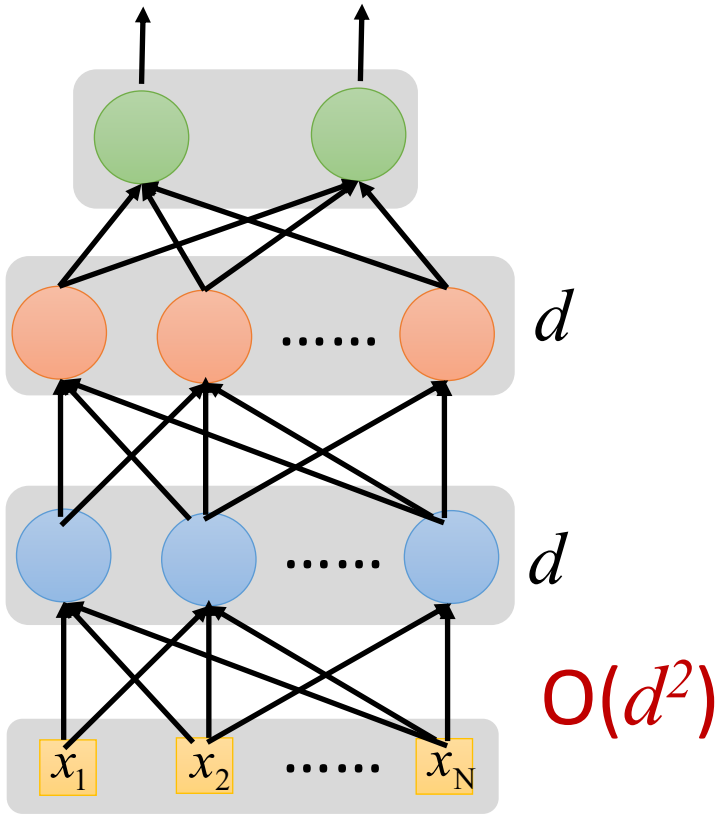
Why “deep” not “wide”?



Fat + Shallow vs. Narrow + Deep

- Two networks with the same number of parameters





How to frame the learning problem?

- The learning algorithm f is to map the input domain X into the output domain Y

$$f : X \rightarrow Y$$

- Input domain: word, word sequence, audio signal, click logs
- Output domain: single label, sequence tags, tree structure, probability distribution

Output Domain – Classification

- Sentiment Analysis

"I loved this product!" → +

"The screen is too small, design is awful" → -

- Speech Phoneme Recognition

→ /h/

- Handwriting Recognition

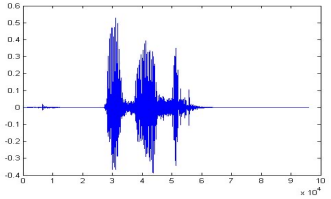
→ 2

Output Domain – Sequence Prediction

- POS Tagging

"I read the book" → I/PN read/VDB the/DT book/NN

- Speech Recognition



→ "the book"

- Machine Translation

"How are you doing today?" → "Bugün nasılsın?"

Learning tasks are decided by the output domains!

Input Domain – How to aggregate information?

- Input: word sequence, image pixels, audio signal, click logs
- Properties: continuity, temporal, importance distribution
- Example
  - CNN (convolutional neural network): local connections, shared weights, pooling
    - AlexNet, VGGNet, etc.
  - RNN (recurrent neural network): temporal information

Network architectures should also consider the input domain properties!

How to Frame the Learning Problem?

- The learning algorithm f aims to map the input domain X into the output domain Y

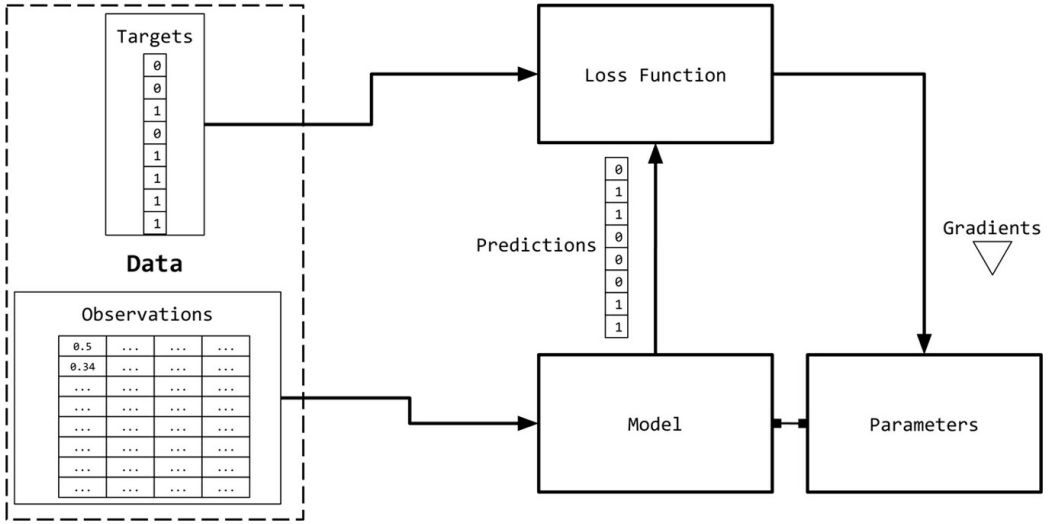
$$f : X \rightarrow Y$$

- Input domain: word, word sequence, audio signal, click logs
- Output domain: single label, sequence tags, tree structure, probability distribution

Network design should leverage input and output domain properties

Observation and Target Encoding

- Representations as numerical vectors.
- Learned from data
- Simple ways to represent:
  - One-hot encoding
  - Term frequency (TF) and bag-of-words
  - Term frequency – inverse document frequency (TF-IDF)



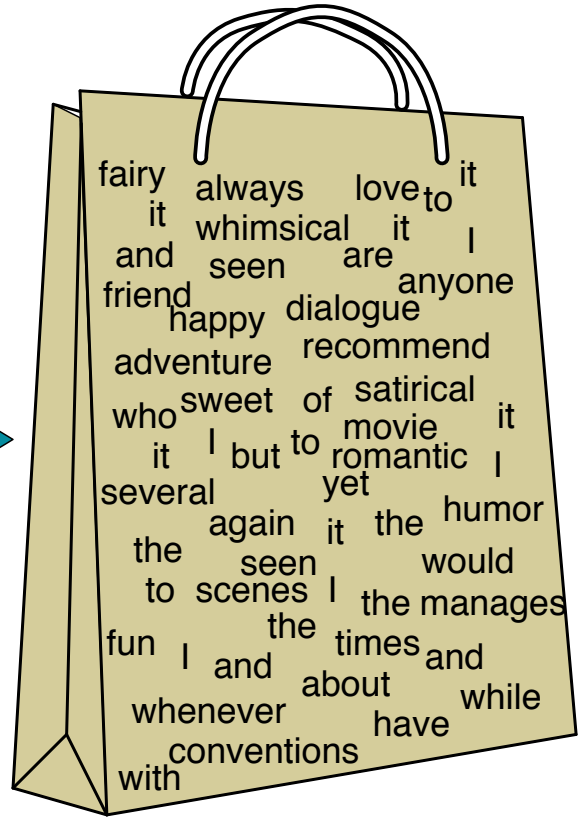
One-hot encoding

- Vocabulary = { time, fruit, flies, like, a , an , arrow, banana}

- ”Time”: [1 0 0 0 0 0 0 0]
- “flies”: [0 0 1 0 0 0 0 0]
- “like”: [0 0 0 1 0 0 0 0]

The Bag of Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
…	…

The Bag of Words Representation

- Vocabulary = { time, fruit, flies, like, a , an , arrow, banana}

…

- ”Time flies like an arrow.”: [1 0 1 1 0 1 1 0]
- “Fruit flies like a banana.”: [0 1 1 1 1 0 0 1]

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Definition of Probability

- Probability theory encodes our **knowledge or belief** about the **collective likelihood of the outcome of an event.**
- We use probability theory to try to predict **which outcome will occur for a given event.**

- For language modeling, we want to calculate the probability of particular **word sequences**, in the most general case.

P(“orange”) or P(”The cat chased the dog.”)

Sample Spaces

- We think about the “sample space” as being set of all possible outcomes:
  - For tossing a coin, the possible outcomes are **Heads** or **Tails.**
  - For competing in the Olympics, the set of outcomes for a given contest are {**gold, silver, bronze, no_award**}.
  - For computing part-of-speech, the set of outcomes are {**JJ, DT, NN, RB, … etc**.}

- We use probability theory to try to predict which outcome will occur for a given event.

The classic example: Flipping a coin

- We flip a fair coin
- What are the possible outcomes (sample space)?
  - Heads (H) or Tails (T)
  - Either is equally likely

- What are the chances of getting H?
  - One out of two
  - P(H) = ½ = 0.5

Another Example

- We roll a die. What are the chances of getting an even number?
- There are six possible outcomes from rolling a die, each with a 1 out of 6 chance: {1, 2, 3, 4, 5, 6}

- There are 3 simple outcomes of interest making up the compound event of interest:
  - Even numbers: {2, 4, 6}
  - Any of these qualifies as “success”
  - Effectively, we can be successful 3 times out of 6

    P(Even) = 3/6 = 0.5

Some Notation…

- We view events and outcomes as sets

- We use Ω (Omega) to denote the total set of outcomes, our event space

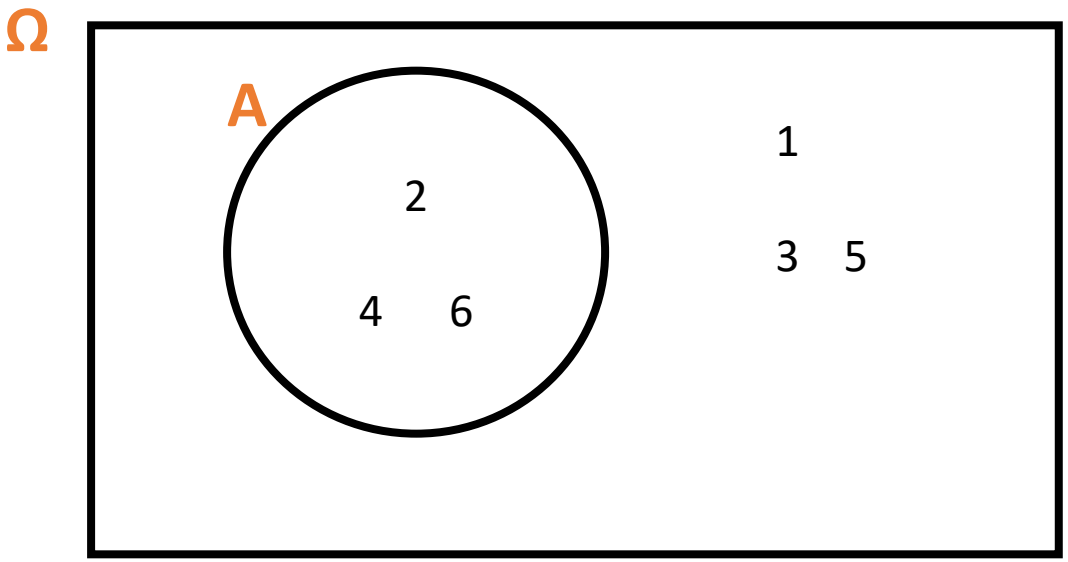
Venn diagram representation of die-throw

- Possible outcomes:

$$\Omega = \{1,2,3,4,5,6\}$$

- Outcomes of interest (denoted A):

$$A = \{2,4,6\}$$



Probability: classical interpretation

- Given n equally possible outcomes, and m events of interest, the probability that one of the m events occurs is m/n.
  - If we call our set of events of interest A, then:

$$P(A) = \frac{|A|}{|\Omega|}$$

Number of events of interest (A) over total number of events

- **Principle of insufficient reason (Laplace):**
  - We should assume that events are equally likely, unless there is good reason to believe they are not.

Compound vs. simple events

- If A is a compound event, then P(A) is the sum of the probabilities of the simple events making it up:

$$P(A) = \sum_{a \in A} P(a)$$

The sum of probabilities, for all elements a of the set A

- Recall, that P(Even) = 3/6 = 0.5
- In a throw of the Dice, the simple events are {1,2,3,4,5,6}, each with probability 1/6
- P(Even) = P(2) + P(4) + P(6) = 1/6 * 3 = 0.5

More rules…

- Since, for any compound event A:

$$P(A) = \sum_{a \in A} P(a)$$

and the probability of all events, P(Ω) is:

$$P(\Omega) = \sum_{e \in \Omega} P(e) = 1$$

(this is the likelihood of “anything happening”, which is always 100% certain)

More rules… (cont.)

- If A is any event, the probability that A **does not** occur is the probability of the complement of A:

$$P(\overline{A}) = 1 - P(A)$$

i.e. the likelihood that anything which is not in A happens.

- Impossible events are those which are not in Ω. They have probability of 0.
- For any event A:

$$P(A) \in [0,1]$$

Example: Throwing a dice

- A = {4}
- Probability that A does not occur
  - complement of A is B = {1, 2, 3, 5, 6}
  - P(B) = 5/6
  - P(A) = 1/6
  - P(B) = 1 – P(A)

- Probability that the outcome of rolling a dice is 8
  - Impossible!
  - P ({8}) = 0

More complex outcomes…

- Here's an even more complicated example:
  - You flip a coin twice.
  - Possible outcomes (order irrelevant):
    - 2 heads (HH)
    - 1 head, 1 tail (HT)
    - 2 tails (TT)

Only one way to obtain this: both throws give H

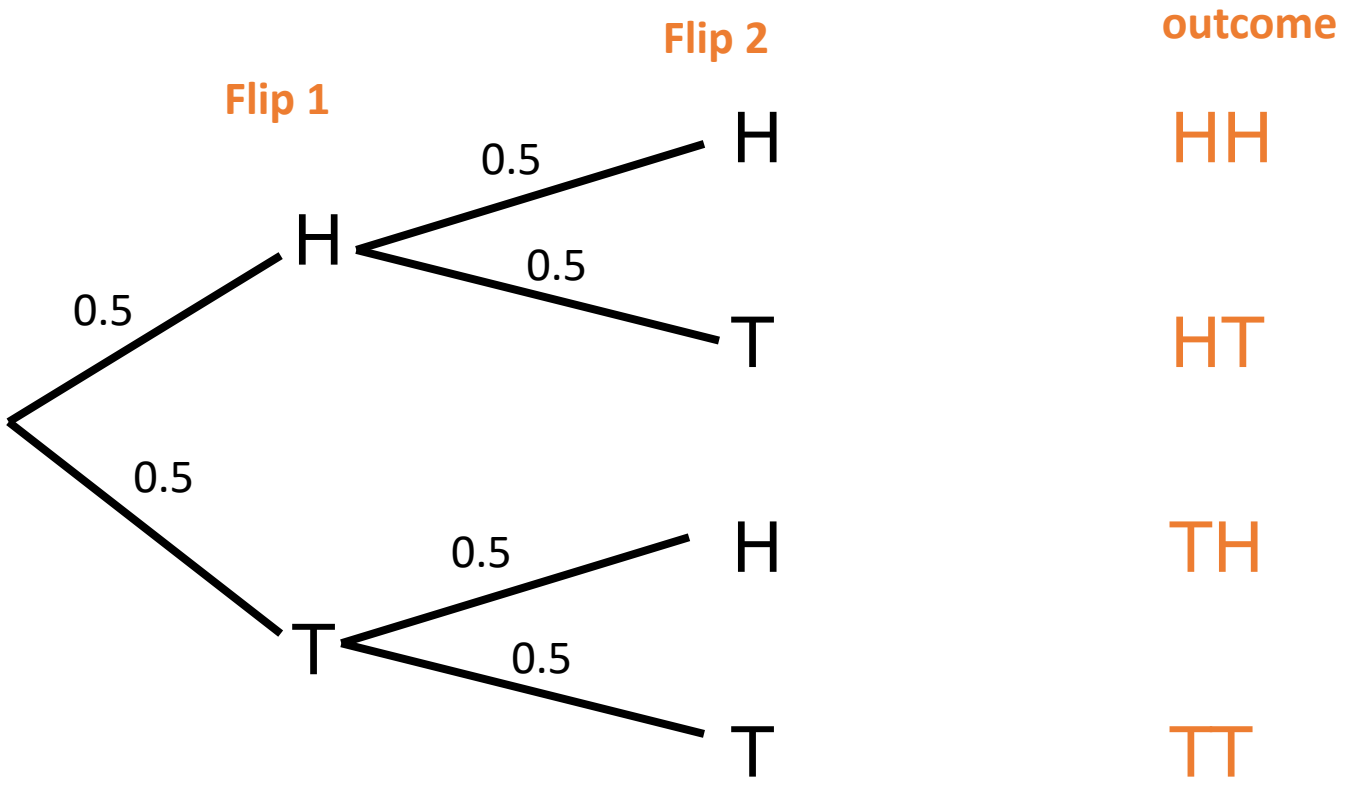
Two different ways to obtain this: {throw1=H, throw2=T} OR {throw1=T, throw2=H}

Only one way to obtain this: both throws give T

- Are they equally likely?
  - No!

Probability Trees

- **Tree diagram** may be used to represent a probability space
- Four equally likely outcomes:



So the answer to our problem

- There are actually 4 equally likely outcomes when you flip a coin twice.
  - HH, HT, TH, TT

- What's the probability of getting 2 heads?
  - P(HH) = ¼ = 0.25

- What's the probability of getting head and tail?
  - P(HT OR TH) = 2/4 = 0.5

Teaser: violations of Laplace's principle

- You randomly pick out a word from a corpus containing 1000 words of English text.
- Are the following equally likely?
  - Word will contain the letter *e*
  - Word will contain the letter *h*
- What about?
  - Word will be *audacity*
  - Word will be *the*

Teaser: violations of Laplace's principle

- You randomly pick out a word from a corpus containing 1000 words of English text.
- Are the following equally likely?
  - Word will contain the letter *e*
  - Word will contain the letter *h*

"e" is the most frequent letter in English orthography

- What about?
  - Word will be *audacity*
  - Word will be *the*

The is far more frequent than *audacity*

- In both cases, prior knowledge or experience gives good reason for assuming unequal likelihood of outcomes.

Unequal likelihoods

- When the Laplace's Principle is violated, how do we estimate probability?
  - We often need to rely on prior experience
  - In general, for language events, P is unknown
  - We need to estimate P
  - By looking at evidence about what P must be based on a sample of data
- Example:
  - In a big corpus, **count** the frequency of *e* and *h*
  - Take a big corpus, **count** the frequency of *audacity* vs. *the*
  - Use these estimates to predict the probability on a new 1000-word sample.

Language Modeling!

Example

- Suppose that, in a corpus of 1 million words:
  - C(*the*) = 50,000
  - C(*audacity*) = 2

- Based on frequency, we estimate probability of each outcome of interest:
  - frequency / total
  - P(*the*) = 50,000/1,000,000 = 0.05
  - P(*audacity*) = 2/1,000,000 = 0.000002

Interpretation of probability

- The core assumption in statistical NLP, where we estimate probabilities based on frequency in corpora:

  Given that a certain event of interest occurs ***m*** times in **n** identical situations, its probability is **m/n**

- Stability of relative frequency:
  - we tend to find that if ***n*** is large enough, the relative frequency of an event ***m*** is quite stable across samples

Corpus Based Statistical NLP

- Based on counting in large corpora and developing different kinds of models
- This approach only became possible with the Web
  - Google Translate: works because it’s seen tons of examples
  - Spelling Correction: also works because it’s seen tons of examples of words in context
  - Auto-complete
  - Word2Vec
- Started out with simple word and POS based approaches, now NLP challenges are more to do with meaning (i.e., GLUE and SuperGLUE).

Outline for Today

- Some Terminology
- What is deep learning?
- Probability
- Conditional Probability and Independence

Prior knowledge

- Sometimes, an estimation of the probability of an event is affected by what is known (which changes the sample space)

Example:

- A box of 10 candies
- **5 red and sweet**, **3 green and sour**, **2 green and sweet**
- You close your eyes and randomly pick a candy, what is the probability that it is sweet?
  - 7/10 = 0.7
- After you pick the candy, you open your eyes and see it is green, now what is the probability that it is sweet?
  - 2/5 = 0.4

Part-of-speech tagging example

- One of the classic tasks in statistical NLP
- Assign a label indicating the grammatical category to every word in a corpus of running text.
- Statistical POS taggers are first trained on data that has been previously annotated. Yields a **language model**.

- Language models vary based on the n-gram window (size of the sequence):
- Unigrams: probability based on tokens (a lexicon)

e.g. input = the_DET tall_ADJ man_NN

model represents the probability that the word *man* is a noun (it could also be a verb)

- Bigrams: probabilities across a span of 2 words

model represents the probability that a DET is followed by an adjective, adjective is followed by a noun, etc.

- Also: Trigrams, 4-grams, etc.

POS tagging

- Suppose we've trained a tagger on the annotated data. It has:
  - A lexicon of unigrams:
    - P(the=DET), P(man=NN), etc
  - A bigram model
    - P(DET is followed by ADJ), etc
  - Assume we've trained it on a large input sample.
  - We now feed it a new phrase:
    - *the audacious alien*
- Our tagger knows that the word *the* is a DET, but it's never seen the other words.
- It can:
  - Make a wild guess (not very useful!)
  - Alternative: **estimate the probability** that *the* is followed by an ADJ, and that an ADJ is followed by a NOUN

Prior knowledge revisited

- Given that I know that *the* is DET, what's the probability that the following word *audacious* is ADJ?
  - This is **very different** from asking what's the probability that *audacious* is ADJ out of context.
  - We have **prior knowledge** that DET has occurred. This can significantly change the estimate of the probability that *audacious* is ADJ.
- We therefore distinguish:
  - **Prior probability**: "Naïve" estimate based on long-run frequency
    - The unconditional probability that is assigned before any relevant evidence is taken into account.
  - **Posterior probability**: probability estimate based on prior knowledge after some observation
    - The **conditional probability** that is assigned after the relevant evidence or background is taken into account.

Conditional Probability

- In our example, we were estimating:
  - P(ADJ|DET) = probability of ADJ given DET
  - P(NN|ADJ) = probability of NN given ADJ
  - …

- In general:
  - The conditional probability P(A|B) is the probability that A occurs, given that we know that B has occurred

Estimation of conditional probability

- Intuition:
  - P(A|B) is a ratio of the chances that both A and B happen, divided by the chances of B happening alone

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

  - P(ADJ|DET) = P(DET+ADJ) / P(DET)

Estimating P(A AND B)

$$P(A \cap B) = P(A)P(B \mid A)$$

Probability that both A and B occur

Probability of A happening overall

Probability of B happening given that A has happened

Multiplication rule

Multiplication rule: example

- We have a standard deck of 52 cards
- What's the probability of pulling out two aces in a row?
  - Standard deck has 4 aces
- Let A1 stand for "an ace on the first pick", A2 for "an ace on the second pick"
- We're interested in P(A1 AND A2)

Example continued

- P(A1 AND A2) = P(A1)P(A2|A1)
- P(A1) = 4/52
  - (since there are 4 aces in a 52-card pack)

- If we do pick an ace on the first pick, then we diminish the odds of picking a second ace (there are now 3 aces left in a 51-card pack).
  - P(A2|A1) = 3/51

- Overall: P(A1 AND A2) = (4/52) (3/51) = .0045
  - in the event of A1, the chances of A2 are diminished
  - the multiplication rule takes this into account

Some observations

- In this example, the two events are not independent of each other
  - occurrence of one affects likelihood of the other
  - e.g. drawing an ace first diminishes the likelihood of drawing a second ace
  - Sampling without replacement

  - If we put the ace back into the pack after we've drawn it, then we have sampling with replacement
    - In this case, the probability of one event doesn't affect the probability of the other.

Independence

- Multiple coin tosses
- Card picks from two different decks (or sampling with replacement)

$$\Pr(x, y) = \Pr(x) \cdot \Pr(y)$$

Dependence

- Picking cards consecutively from a deck (i.e., sampling without replacement)
- Message exchanges between two friends

$$\Pr(x, y) \neq \Pr(x) \cdot \Pr(y)$$

Extending the multiplication rule

- The logic of the "A AND B" rule is:
  - Both conditions, A and B have to be met
  - A is met a fraction of the time
  - B is met a fraction of the times that A is met
- Can be extended indefinitely
  - E.g. chances of drawing 4 straight aces from a pack
  - P(A1 & A2 & A3 & A4)

$$= P(A1)\ P(A2|A1)\ P(A3|A1\ \&\ A2)\ P(A4|A1\ \&\ A2\ \&\ A3)$$

Bayes Rule

- Joint Probability

$$\Pr(X, Y) = \Pr(X|Y)\Pr(Y) = \Pr(Y|X)\Pr(X)$$

- Bayes Rule

$$\boxed{\Pr(X|Y) = \frac{\Pr(Y|X) \cdot \Pr(X)}{\Pr(Y)}}$$